Contextualization and Knowledge Graph for Plant DDN-01

Industrial AI take-home assignment – report

AKASH November 2026

Result in one paragraph. A Python pipeline links the siloed records of plant DDN-01 to one asset hierarchy and loads them into a Neo4j knowledge graph with provenance on every edge. It runs end to end with one command (`run_all.bat` on Windows, `make all` elsewhere; 18 steps, 102 tests). Tags are matched with **F1 1.000** and work orders with **F1 0.982** (out-of-fold on the TRAIN labels; the hidden test set is the real measure), with **0 wrong auto-accepted links**. The graph holds 165,706 nodes and 1.14 M relationships, answers the 7 required queries, contains 8 anomaly events (all 8 seeded signal problems found, 1 judged a false alarm), and supports a GraphRAG question box that scored **10/10 on its evaluation set and 4/5 on unseen questions** (manually reviewed) and never stated an invented fact in any run. All data is the synthetic DDN-01 pack.

1. Part A – How industrial platforms do contextualization

Compared: **Cognite Data Fusion (CDF)**, **Siemens** (Insights Hub, Industrial Information Hub, Industrial Copilot) and **AWS IoT TwinMaker**, from vendor documentation. The full version with every link is `docs/part_a_research.md`.

> Every link was opened and each claim checked against the live vendor page (November 2026). ⚠ marks a claim searched for but not found on an official page.

How the plant is modelled. CDF's *Data Modeling Service* separates **containers** (physical storage of properties), **views** (logical schemas over containers, which can rename, combine and inherit) and **data models** (groups of views), all scoped to a *space* (Cognite: containers, views, data models). Cognite ships a **Core Data Model** (`cdf_cdm`) with `CogniteAsset`, `CogniteEquipment`, `CogniteTimeSeries`, `CogniteFile`, `CogniteActivity`; `CogniteAsset` carries parent, root and an automatic path (Core data model). Siemens **Insights Hub** builds *asset types* from *variables* and *aspects*, inherited from the parent type (Asset Management); **IIH Semantics** exposes models and time series through a GraphQL API backed by a knowledge graph (IIH). AAS support is a separate product, **SITRANS assetfox** (assetfox) ⚠ (none found inside Insights Hub). In **TwinMaker**, twins are *entities* with *components*; relationship properties become edges of its knowledge graph and parent links add `isChildOf` (TwinMaker knowledge graph).

How records are linked.

Method	Cognite	Siemens	AWS
Rules	Regex rules and confirmed matches are applied **before** model predictions (matching)	Configuration in the asset model and connectors ⚠	Declared on components
Fuzzy / ML	Unsupervised field-similarity model by default; **supervised** when `trueMatches` labels exist (entity matching API)	No documented ML matcher ⚠	None built in
Diagrams	Tags, symbols and connections from P&IDs; raster gives tags only (diagram parsing)	–	–
Review	Quick match, pipelines, manual confirmation (match entities)	–	–

How confidence is handled. Cognite is the most explicit. A pipeline uses a model match only above a score threshold and only if it is not in the rejected list; confirmed matches take priority and `refit` retrains on them (matching, SDK). `CogniteAnnotation` has a `confidence` and a `status` of *Suggested / Approved / Rejected* (core model). No equivalent predicted-link workflow was found for Siemens or AWS ⚠: their links are configured, not predicted.

Standards. ISA-95 hierarchies appear in all three (`CogniteAsset` path (hierarchy guide), Insights Hub asset tree, TwinMaker `isChildOf`). Siemens Rapidminer **Graph Studio** uses W3C ontologies (Graph Studio); Cognite and AWS use their own property-graph models. Cognite binds units from a catalog at container level (units). OPC UA ingestion is via extractors (Cognite) ⚠, native (Siemens), and SiteWise (AWS) ⚠.

How GenAI uses the graph. Cognite Atlas AI agents call tools such as "query knowledge graph", document Q&A and time-series analysis; they require the Core Data Model and are separately licensed (Atlas AI, agent tools). **Siemens Industrial Copilot** (with Microsoft, Azure OpenAI) generates automation code and helps diagnose faults; Schaeffler uses it on the shop floor (Siemens press). Its grounding is engineering data more than a plant graph ⚠. **TwinMaker** queries its graph with PartiQL `MATCH`, including variable-length paths, which an agent can call as a tool (queries).

Values vs metadata. All three keep raw values out of the graph. A `CogniteTimeSeries` is a node, but its datapoints go through the time-series API (time series in DM). Siemens stores records per asset + aspect in the IoT Time Series service (IoT Time Series). TwinMaker components point at SiteWise or Timestream through connectors.

Limits.

	Cognite	Siemens	AWS
Lock-in	Proprietary DMS schema; migration means re-modelling	Tied to the Siemens automation stack	Tied to SiteWise / Timestream
Cost	Atlas AI licensed separately	Subscription	Knowledge graph charged per query (pricing)
Operations	A changed container means a new container and re-ingest	Ingest throttled per asset/aspect	–
Still needs people	Confirming matches, writing rules, fixing tag misreads (troubleshooting)	Building types and aspects	Declaring every relationship

Automotive use cases. Siemens: Schaeffler (link above, press release). AWS: Volkswagen Digital Production Platform, 43 sites, GenAI maintenance assistant (AWS case study); it does not state that TwinMaker is used. Cognite: no named automotive OEM case found ⚠; closest is Aarbakke, machining with welding robots (case).

Ideas borrowed for Parts C and D. From **Cognite**: rules and engineer-confirmed matches before the ML model; a score threshold plus a remembered rejected list; `confidence` + `status` (suggested / approved / rejected) on every link; confirmed links reused as training labels (`refit`); and the **container / view** split, mirrored as one storage schema with role-specific views in the ontology file. From **Siemens**: equipment *types* that define which signals a type should have, used as the "unit fits equipment type" feature. From **AWS**: variable-length path patterns for the `feeds` traversal (query D4). From all three: values stay outside the graph, which holds only pointers.

2. Part B – What the data pack contains

14 files were profiled (`reports/01`, `notebooks/01_data_profiling.ipynb`): 455 assets, 1,541 tags in four naming conventions, 1.1 M one-minute values for 55 tags, 7,200 VINs with 310,462 RFID passages, 79,855 torque results, 64,800 part-lot records, 675 defects, 2,968 work orders and 7,917 alarms.

Data quality. 38 issues, each with a count and verifiable examples (`reports/02`); **15 are not in the data dictionary**. Examples: 18 assets with a missing parent, a relocated robot whose tag keeps the old name, °F and °C in the same historian, a stuck sensor, 815 alarms writing `S3` for station `ST03`, SAP failure codes that contradict the work-order text, and a supplier lot id reused across two fitting sessions.

Identity schemes. The canonical key is the SAP **functional location** (`DDN1-BIW-L2-ST04-RB07`). The same robot is `DDN1_BIW_L2_R07_WELDCUR.PV` in PI, `RB-207` on the drawing, `B2-04` in MES and "R7" or "robo 7" in Hinglish work orders (`reports/03`). Defects are logged against VIN and inspection point, never against equipment, so only the graph can connect a defect to a machine.

Production problems found (`reports/04`; method: anomaly window → VINs exposed at that equipment → defect rate vs. all other VINs):

#	Problem	Exposed VINs	Defective exposed / not	Lift
1	Press line 1 die set-up error → split body-side panels	300	28 / 0	all
2	Nutrunner NR14 transducer offset → low wheel-nut torque	37	37 / 0	all
3	Robot L2 R07 weld-current drift → cold welds	634	101 / 6	174×
4	Door-seal lot WS-260904-B → water leaks (lot id reused)	1,492	85 / 0	all
5	Sealer SR03 heater failure → roof-ditch leaks	106	36 / 1	2,409×
6	Booth B humidifier valve stuck open → paint craters	29	7 / 0	all
7	Oven 2 zone 3 over-temperature (no defects: false alarm)	19	0 / 81	0×
8	AHU TC1 supply-fan bearing wear (caught before failure)	–	–	–
9	E-coat TT-301 transmitter frozen (sensor fault)	–	–	–

3. Design decisions

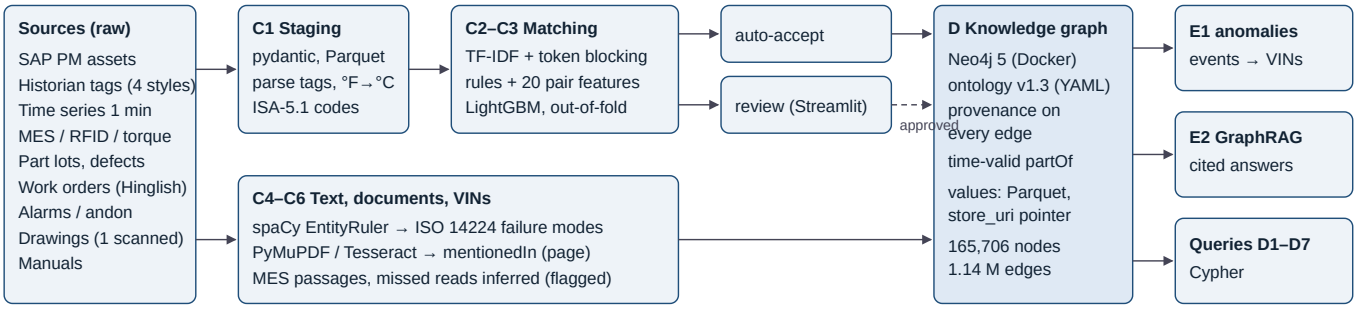


Figure 1. Pipeline as built. Only auto-accepted or engineer-approved links enter the graph.

Canonical key: the SAP functional location. It is the only identifier every engineer recognises, it encodes the ISA-95 path (site–shop–line–station–equipment–component), and it survives renaming in the historian. Every other identifier (tag, MES station, drawing label, alias in work-order text) becomes either a property (`aliases`, `mes_code`) or a linked node.

Matching (Part C). Comparing every record with every asset (1,541 × 455) is wasteful and gets worse at scale, so **blocking** keeps the top-15 assets by character n-gram TF-IDF plus a token index on equipment codes. That is ~25 candidates per record instead of 375, with blocking recall 1.000 (tags) and 0.999 (work orders). A rules matcher (exact code, alias, relocation, SAP equipment number) is kept as a baseline *and* as a feature. For each pair, 20 features (rule agreement, fuzzy name/FL scores, line and component conflicts, unit fits equipment type, SAP equipment number agrees with the text, valid at the record's time) feed **LightGBM**. Scores are **out-of-fold** with GroupKFold by record, so no record is scored by a model that saw its label.

Thresholds by cost, not by F1. In a recall investigation, a wrong auto-accepted link sends engineers to the wrong machine, which is worse than a link waiting for review. The auto-accept threshold is therefore the lowest score on the out-of-fold precision curve with **precision ≥ 0.995 and no wrong-machine error** (0.932 for work orders, 0.50 for tags). Below it there are **three bands, as in the assignment's Figure 1**: **review** (the Streamlit review screen) and, below a second threshold, a **backlog** (data-quality queue). The backlog cut-off is also read from the out-of-fold data: the highest score under which the top candidate was right at most 25% of the time. For work orders that is **0.093** (of the 6 labelled orders below it, 1 was right), so 52 orders go to the backlog. For tags there is **no backlog**: even the lowest-scoring labelled tags were right (10 of 10), so the data does not support one. Two exceptions keep the backlog from hiding anything important. A backlog work order that **explains a detected anomaly** is pulled into review and shown first (WO 4102961, the robot R7 repair, scores 0.034). And every other backlog record is offered to review with **link prediction's top 3** (Part E3, below). Each decision is stored in `data/review/decisions.csv`, becomes a `method=engineer` edge, and is reused as a training label on the next run.

Text and documents. Work orders: a Hinglish normaliser ("robo", "nahi chal raha"), then a spaCy EntityRuler for equipment mentions and a phrase dictionary for **ISO 14224** failure modes (`STD`, `ERO`, `FTS`, …). Drawings:

PyMuPDF text layer, or **Tesseract OCR** for the scanned press-line drawing. OCR confidence is lower (0.85, or 0.70 after a correction, which goes to review). Labels like `RB-207` are resolved with the drawing's own line context.

VIN traceability. MES stations are mapped to FLs (46/46). Missed RFID reads (6,332, 2%) are filled from line sequence and takt time (2.4 min) and flagged `inferred=true`. The inference was validated by hiding 5,793 real reads: the median error is 0.28 min, and 95.7% are within one takt. The large errors sit right after buffers, so each inferred passage stores `uncertainty_min` and gets a lower confidence.

Ontology first; property graph. `ontology/ontology_v1.yaml` (v1.3.0, 18 node types, 19 edge types) is the single source of truth. It is validated with pydantic, rendered to `docs/ontology.md`, and used both by the loader (which rejects edges whose endpoints the ontology forbids) and by GraphRAG (the schema the LLM sees). Following Cognite, it separates **containers** (what is stored: labels + properties + indexes) from **views** (Maintenance, Quality, Traceability: what each role sees). I chose a **property graph (Neo4j Community + Cypher)** over RDF. The questions are path questions with properties on the edges (time, confidence, provenance), and that is native in a property graph but needs reification in RDF. Cypher is easier to validate for LLM-generated queries, and it maps directly onto Cognite DMS (nodes, edges with properties, spaces). RDF/OWL would win for cross-company semantic integration (AAS, supplier data).

Provenance and time. Every edge has `source_system`, `method` (source_key / rule / fuzzy / ml / engineer / inferred / derived), `confidence`, `created_at` and `status`, plus `evidence` where useful ("rules (exact) and ML agree | ML p=1.0"). Robot RB12 moved from Line 1 to Line 2 in March 2026. `partOf` has `valid_from/valid_to`, so a VIN built before the move still traces to the old line. Time-series **values stay in Parquet**; the `TimeSeries` node holds metadata, the latest value and a `store_uri`.

Keeping it fast. 7,200 VINs × ~40 stations = 316,794 `passedThrough` and 511,182 `processedBy` edges. The load uses batched `UNWIND` writes with uniqueness constraints and indexes on keys. Queries always start from an indexed key (VIN, lot, FL), never from a label scan. At the assignment's scale (18,000 VINs) this stays in seconds. Section 6 covers what changes at 2 M VINs a year.

Anomalies (E1). No hand-set limits. Each signal is compared with its own lagged 3-day median/MAD while the line is producing (a robust z-score), with a fast (12 h lag) and a slow (1 day lag) horizon, hysteresis (start $|z| > 6$, continue > 3) and ≥ 10 min duration. There is also a flat-line detector and a per-tool torque z-score. **Isolation Forest** was run as a comparison only: it flagged more windows with no clear meaning. Each event is written as an `AnomalyEvent` on its equipment and tag, with `exposedTo` edges to the VINs processed in the window. It is then **judged** from context: the defect lift on exposed VINs, a confirming work order, and the bake window from the oven manual.

GraphRAG (E2). The flow is: question → LLM writes Cypher from the ontology schema → **validator** → **read-only** transaction with a timeout → rows (+ fixed-query evidence expansion) → LLM answers as sentences with citations → **every citation must appear in the rows**, otherwise the sentence is dropped. The validator rejects write keywords, unknown labels, relationships and properties, wrong arrow directions, and one variable reused for two node types, and it checks the query with `EXPLAIN`. Three paths lead to "I don't know": the model says the schema can't answer, the query returns 0 rows, or no sentence survives grounding. LLM: **OpenAI gpt-4o** (temperature 0, structured JSON output), because a key was available and it was the strongest model tested. Claude and Ollama are switchable, and an offline template mode keeps the pipeline and tests running without any key.

Two records end to end

One robot tag: raw name → graph node (`python -m ddn.demo tag`):

Step	Result
Raw (PI historian)	`DDN1_BIW_L2_R07_WELDCUR.PV`, unit kA, "Robot 07 Weld current"
Parsed	shop BIW, line L2, equipment R07 → **RB07** (PI writes R07, SAP RB07), signal WELDCUR → ISA-5.1 `IT`
Matched	`DDN1-BIW-L2-ST04-RB07`, ML p = 1.000, band auto-accept (the station ST04 is not in the tag; it comes from the asset master)

Step	Result
Graph edge	`(RB07)-[:hasTimeSeries {source_system: PIPELINE, method: rule, confidence: 1.0, status: approved, evidence: "rules (exact) and ML agree"}]->(TimeSeries)`, values at `store_uri = parquet:…/tag_name=DDN1_BIW_L2_R07_WELDCUR.PV`
Context	partOf RB07 → ST04 → L2 → BIW → DDN1; anomaly AN-003 (7–8 Sep), confirmed by WO 4102961; 13 work orders on the robot

One work order: raw text → edge, and where it can go wrong. WO 4102961: *"R7 gun tip dressing fail, cap changed" / "Weld current low aa raha tha since yesterday. L2 ST4 robot 7. Tip dresser blade chipped, caps changed, current back to 9.4 kA."* There is no SAP equipment number. The normaliser turns the Hinglish into "current low", "ST4" into ST04 and "robot 7" / "R7" into RB07, giving the tokens L2, ST04, RB07 plus the components WG (gun) and TD (tip dresser). The failure-mode step gives ISO 14224 `LOO` (low output; also SER, STD). The matcher's top candidate is the **weld gun** `DDN1-BIW-L2-ST04-RB07-WG`, at a low score. This is where it goes wrong: the text names the robot, its gun *and* the tip dresser, and SAP could hold the order on any of the three. Its score, 0.034, is even below the backlog cut-off. But it **explains anomaly AN-003**, so it is pulled into the review screen and shown first. Once an engineer approves or corrects it, it becomes a `method=engineer` edge and a new training label.

One VIN end to end (`python -m ddn.demo vin`): MDDA5XK00D003924 (Hatch 1.2 MT) entered the body shop on 8 Sep at 08:55. It passed 44 stations to the end-of-line audit at 20:00 and was processed by 40 weld robots, 6 sealer robots and 10 paint robots. It has 11 torque results (all OK, tools TC10–TC15) and 9 parts with their lots, including door seal WS-260904-B. Its one defect, D700356 *BW-017 weld nugget small*, found at body inspection, is `causedBy` AN-003 on robot `DDN1-BIW-L2-ST04-RB07` (derived, confidence 0.9). That anomaly was fixed by WO 4102961. Every line of that trace is one Cypher query result.

4. Metrics

Entity matching (out-of-fold on the TRAIN labels: 539 tags, 888 work orders; `reports/05`, `06`):

Source	Method	Precision	Recall	F1	Wrong links
Tags	rules only	1.000	0.974	0.987	0
Tags	ML (top candidate)	1.000	1.000	**1.000**	0
Tags	ML, auto-accept band only	1.000	0.981	–	0
Work orders	rules only	0.976	0.925	0.950	20
Work orders	ML (top candidate)	0.982	0.982	**0.982**	16
Work orders	ML, auto-accept band only	**1.000**	0.957	–	**0**

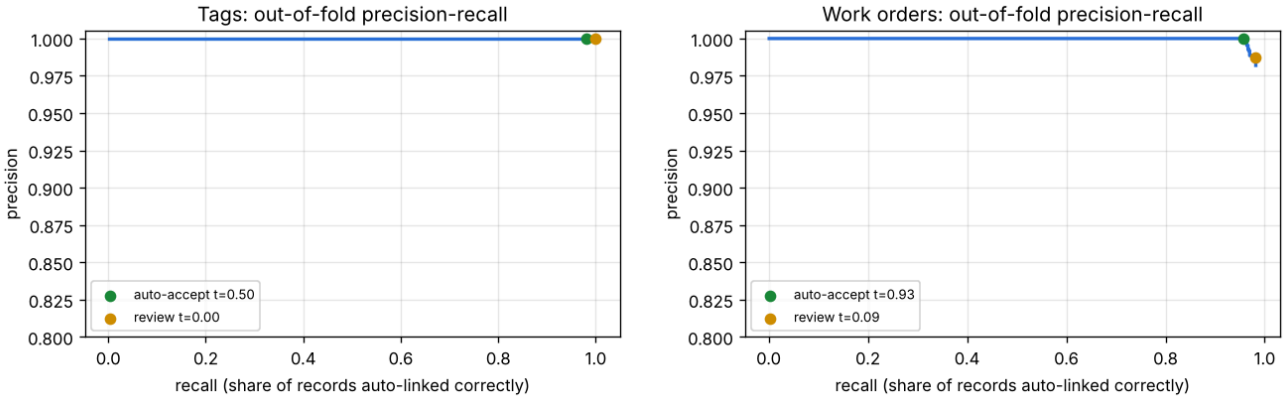


Figure 2. Out-of-fold precision–recall curves, tags (left) and work orders (right), with the chosen auto-accept point.

Per source system and per rule (tags, rules baseline). The ML matcher reaches F1 1.000 on every source system, so this table shows where the rules alone lose recall, and therefore what the ML layer has to fix:

Group	n	Rules F1	Rules misses	Group	n	Rules F1
PI Historian	381	0.989	8	rule: exact	495	1.000

Group	n	Rules F1	Rules misses	Group	n	Rules F1
S7 legacy OPC	69	0.970	4	rule: alias	24	1.000
MES tags	55	0.981	2	rule: relocation	6	1.000
WinCC SCADA	34	1.000	0	no rule fired	14	0.000

Rule precision is 1.000 everywhere: a rule that fires is right, and its only weakness is recall.

Where each method fails. Rules miss **typos and unusual formats**: `DD1_` instead of `DDN1_`, glued sub-units like `PR2LUB`, and Hinglish text with no code. ML fixed 14 tag and 51 work-order records the rules got wrong or missed, and broke none. ML's remaining errors are **ambiguity**, not noise. "RB01 gun water leak" with no line could be robot 1 on L1, L2 or L3. Errors also come from robot vs. its weld gun ("R12 ka gun nahi chal raha": robot or gun?) and from a few contradictory labels. Most sit below the auto threshold and are caught by review.

Coverage and review load. Contextualized automatically: **99.4% of tags** (1,532 / 1,541) and **95.9% of work orders** (2,846 / 2,968). Review queue: 9 tags, 70 review-band work orders + 1 pulled up from the backlog because it explains an anomaly, 51 backlog work orders with link-prediction suggestions, and 44 document mentions. Backlog: 52 work orders, 0 tags. Document mentions: 255 found, all resolved, 211 auto-accepted (≥ 0.90). Work orders: 100% of corrective work orders received an ISO 14224 failure mode. Hidden-test predictions are written to `outputs/predictions_*.csv` (1,002 tags and 2,080 work orders not in TRAIN), with confidence and band.

Graph. 165,706 nodes, 1,137,011 relationships, loaded in about 3 minutes. Relationships without full provenance: **0**; endpoints violating the ontology: **0**; provenance values outside the allowed lists: **0**. All 7 required queries return correct results (`reports/12`). Highlights: D1 lists the 4 Oven-2 temperature sensors with latest values converted from °F. D3 shows that NR14 had **no calibration work order in 12 months**, one of only 2 of 20 nutrunners. D5 shows the lot-id recall scope of **1,785 vehicles** shrinks to **1,492** when fitting time is used, which saves 293 good cars from a recall. D7 (plant head): 1,492 + 647 + 106 + 37 cars carried a known risk, and inspection caught only part of them.

Anomalies. 8 events: **all 8 seeded signal problems found**, each written into the graph on its equipment, with `exposedTo` edges to the VINs processed in the window (`reports/13`):

Event	Equipment	Window	VINs exposed	Top defect on exposed VINs	WO	Verdict
AN-001	Press PL1-PR1	02 Sep 10:04–16:34	300	BP-031, every case (28)	4102965	quality
AN-002	Nutrunner NR14	03 Sep 08:30 – 04 Sep 10:40	646	AU-077, every case (38)	4102964	quality
AN-003	Robot L2 RB07	07 Sep 12:45 – 08 Sep 15:14	663	BW-017, 254× (103)	4102961	quality
AN-004	Sealer SR03	09 Sep 06:08–10:24	108	WL-101, 2,364× (36)	4102962	quality
AN-005	AHU TC1 supply fan	09 Sep 14:11 – 12 Sep 16:50	956	PD-007, 2.4× (4)	4102967	equipment
AN-006	AHU TC2	10 Sep 13:04–15:34	27	PD-020, every case (7)	4102963	quality
AN-007	E-coat TT-301	11 Sep 00:00–09:29	237	–	4102968	sensor fault
AN-008	Oven 2 zone 3	12 Sep 02:04–03:34	19	–	4102966	**false alarm**

Reasons for the three that are not quality problems. AN-005: vibration rose for 3 days but defects barely moved (2.4×), so it is an equipment problem caught before failure. AN-007: the signal flat-lined while the line was running, which is a data problem, not a process problem. AN-008: a real over-temperature, but under 2 h and under +30 °C, inside the bake window from the oven manual, and the cars in the oven had no

extra defects. Problem 4 (the supplier lot) has no signal; it is found through genealogy and loaded as a `causedBy → SupplierLot` hypothesis (`status=suggested`).

GraphRAG. Scores improved run by run as the evaluation exposed problems:

Run	Change	Tuned 10	Held-out
1	gpt-4o-mini, schema-only prompt	2/10	–
2	gpt-4o; real relationship patterns, worked examples, direction check, retry on 0 rows	7/10	–
3	stations-of-a-line rule, variable check, citing IDs inside text	8/10	–
4	supplier-lot cause in the graph; evidence expansion	10/10	–
5	first held-out set; temperature 0	8/10	2/5
6	general fixes by failure type; **fresh** held-out set	**10/10**	**4/5** manual (5/5 automatic)

The tuned score is optimistic because the prompt was adjusted on those questions; the **unseen round-2 set** is the fair number. The automatic check only tests that the expected IDs appear in the answer. Reading the answers showed that one held-out answer ("which work order fixed the sealer nozzle problem?") **led with a routine PM order** (4102593) and gave the correct repair, 4102962, second. So it is **4/5 on manual review**. No answer in any run contained an invented fact: every cited value came from the graph. The 10 questions and the held-out set, with expected and actual answers, are in Appendix B.

E3 (bonus): suggesting missing links (`reports/15`). The matcher scores pairs with designed features; link prediction instead learns from the graph. Each asset gets a node embedding: its own tokens plus the mean of the vectors of the records already linked to it, which is one round of GraphSAGE-style mean aggregation with no learned weights. Records are then ranked against all 455 assets by cosine similarity. Out-of-fold (a fold's own links hidden):

	n	Link prediction hit@1	hit@3	hit@5	Matcher top-1 (out-of-fold)
All labelled records	1,427	0.820	–	0.965	0.989
Hard: matcher unsure or wrong	20	0.30	**0.75**	0.85	–

It is weaker than the matcher overall, as expected without designed features. On the hard records its top-1 is unreliable, but its top 3 holds the answer 75% of the time: for 11 of the 16 records the matcher got wrong. So it is used only as a short list for the engineer, never auto-accepted. Examples: WO 4100629 *"Station 04 L2 fixture locating pin worn"*, where the matcher guessed a conveyor and link prediction gives `DDN1-BIW-L2-ST04-FX`; and WO 4100244 *"PL1 DST jam"*, where the matcher guessed the press and link prediction gives the destacker. A first version with a spectral (SVD) embedding reached only hit@1 0.03, because siblings like robot 07 on three lines share their tokens; aggregating the approved links is what separates them.

5. Limitations

- **Synthetic data, TRAIN-only scores.** All matching scores are out-of-fold on the TRAIN labels. The rules were designed after reading those labels, so the hidden test set is the real measure. Tags at F1 1.000 mostly shows that the synthetic naming is regular.
- **Thresholds from little data.** The auto threshold for work orders rests on ~850 auto-linked labelled records; one more error would move it. The backlog cut-off rests on only 6 labelled work orders, and tags have no backlog at all because the labelled data gives no reason for one.
- **Review band holds root causes.** 4 of the 8 root-cause work orders (including NR14's calibration) scored in the review band, so the graph is incomplete until an engineer approves them.
- **Sentence embeddings off.** `sentence-transformers` could not download its model in the build environment, so the embedding feature is empty. The other features made up for it here; on real free text it would matter more.
- **Inferred passages after buffers** can be off by up to 69 min. They are flagged and carry `uncertainty_min`, but a query that ignores these flags can blame the wrong shift.

- **Root causes are hypotheses.** Anomaly verdicts use defect lift over two weeks. The supplier-lot cause needs a defect-to-part map (WL-104 → door seal), because pure statistics also flag every other lot built in the same days.
- **GraphRAG.** It is evaluated on 20 questions, a smoke test rather than a benchmark. Asset-name matching uses exact `CONTAINS` ("TC1 supply fan" ≠ "Supply fan AHU-TC1"). The validator is regex-based (the read transaction is the real write guard). Questions and rows are sent to OpenAI; that is acceptable for synthetic data but not for a real plant without an approved deployment.
- **GraphRAG scoring.** The automatic check passes an answer if the expected IDs appear anywhere in it, so it misses answers that lead with the wrong record. Answers were reviewed by hand; a stricter grader (an LLM judge, or checking the first claim) would be the next step.
- **E3 link prediction** is a parameter-free aggregation, not a trained GNN. Its two settings were picked from 6 combinations on the same labelled folds, so its scores are slightly optimistic. **Not attempted:** the optional seeded data generator.
- **Part A** compares documentation, not hands-on use: no vendor platform was used to build this pipeline.
- **Neo4j Community**: a single instance, no role-based access, no clustering.

6. What changes with real plant data

Scale (10 plants, 50,000 tags, 2 M VINs a year). At 40 stations per car, passages alone reach 80 M edges a year, plus ~130 M `processedBy` edges. I would keep **assets, tags, work orders, defects, lots and anomalies in the graph**, and move the per-station passages to a **partitioned columnar store** (Parquet or Delta by plant and day), exactly like time-series values. The graph then gets only VIN → line-segment edges and the exceptions (inferred reads, rework, anomalies exposed). Each plant becomes its own space or database, with cross-plant queries through a shared asset type and supplier-lot layer. In Cognite, the same split is spaces + DMS for structure and the time-series API for values.

Matching. Each plant has its own naming conventions, so parsing rules become per-plant configuration and the model is retrained per plant with **active learning**: the review queue is ordered by expected value (anomaly-explaining and low-margin items first) so the labels engineers give are the most useful. I would **measure on real data first**: label a random sample of 200 records per source and plant, compare with the synthetic scores, and track precision of auto-accepted links monthly. Drift appears when a new line or PLC naming scheme arrives.

Streaming and MES. Batch files become event streams (OPC UA / MQTT into Kafka). MES would give real start and end times per station, which replaces the takt-time inference, and the graph is updated incrementally instead of reloaded.

Governance. The ontology needs an owner and versioned releases (it already has a changelog). Review decisions need audit and role-based access. The LLM must run on an approved deployment (Azure OpenAI in the company tenant, or a local model), with the same read-only and citation rules.

What still needs people. Confirming ambiguous links, accepting root-cause hypotheses (`status=suggested`), writing defect-to-part maps, and deciding recall scope. The pipeline's job is to make those decisions fast and traceable, not to remove them.

7. Use of AI assistants

An AI coding assistant (Claude Code) was used to draft code, tests and documentation. Every file is listed in `docs/AI_USAGE.md` with what the assistant did. Every result in this report was produced by running the pipeline on the pack, and every number was checked against the generated reports. GraphRAG itself uses OpenAI gpt-4o at run time.

> **Run it:** copy the pack into `data/raw/` and `docs/source_pack/documents/`, then `run_all.bat` (Windows) or `docker compose up -d && make all` (Mac/Linux). The README has details; `docs/LAPTOP_SETUP.md` lists expected outputs per step.

Appendix A – The seven required queries: Cypher and real output

Generated from `reports/12_graph_queries.md` (produced by `python -m ddn.queries` against the loaded graph). Tables are cut to 2 rows; the full output is in that file.

D1. All temperature sensors in Topcoat Oven 2, with latest value and open work orders

View: Maintenance. Parameters: `oven` = DDN1-PNT-OV2, `as_of` = 2026-09-15 00:00:00.

```
MATCH path = (a:Asset)-[:partOf*0..3]->(:Asset {fl: $oven})
WITH a, relationships(path) AS rels WHERE all(x IN rels WHERE x.valid_to IS NULL)
MATCH (a)-[:hasTimeSeries]->(t:TimeSeries {isa51_code: 'TT'})
OPTIONAL MATCH (a)-[:hasWorkOrder]->(w:WorkOrder)
WITH a, t, collect(w) AS wos
RETURN a.fl AS equipment, t.tag AS sensor, t.unit AS unit_in_source, t.unit_canonical AS unit,
       round(t.latest_value, 1) AS latest_value, t.latest_at AS latest_at,
       [w IN wos WHERE w.closed_at IS NULL OR w.closed_at > $as_of | w.wo_number] AS open_work_orders,
       size(wos) AS work_orders_12m
ORDER BY equipment, sensor
```

equipment	sensor	unit_in_source	unit	latest_value	latest_at	open_work_orders	work_orders_12m
DDN1-PNT-OV2-Z1	PaintShop-OVN2-Z1-Temp	degF	degC	139.4	2026-09-14 23:59	9	
DDN1-PNT-OV2-Z2	PaintShop-Ovn02-Z2-Temp	degF	degC	150.2	2026-09-14 23:59	4	

... 2 more rows

- 4 temperature sensors (deg F in WinCC, stored as deg C). Open work orders: 0 – every work order in the export is closed, so the list is empty by data, not by mistake.

D2. A VIN with a water-leak defect: which sealer robot and nozzle processed it, and what were sealer temperature and flow at that time

View: Quality. Parameters: `vin` = MDDA5XK0XD004375.

```
MATCH (v:Vehicle {vin: $vin})-[:hasDefect]->(d:Defect) WHERE d.defect_family = 'water leak'
MATCH (v)-[p:processedBy]->(e:Asset) WHERE e.asset_type IN ['SealerRobot', 'Dispenser']
OPTIONAL MATCH (e)-[:hasTimeSeries]->(t:TimeSeries) WHERE t.store_uri IS NOT NULL
RETURN v.vin AS vin, d.defect_code + ' ' + d.location AS defect, e.fl AS equipment, e.asset_type AS type,
       p.start AS start, p.end AS end, p.inferred AS passage_inferred, p.confidence AS edge_confidence,
       t.tag AS signal, t.store_uri AS store_uri
ORDER BY equipment, signal
```

vin	defect	equipment	type	start	end	passage_inferred	edge_confidence	signal
MDDA5XK0XD004375	WL-101 Roof ditch	DDN1-PNT-SL-ST01-SR01	SealerRobot	2026-09-09 06:17	2026-09-09 06:20	False	0.95	DDN1_PNT_SL_SR01_NOZTEMP.PV
MDDA5XK0XD004375	WL-101 Roof ditch	DDN1-PNT-SL-ST01-SR01-DSP	Dispenser	2026-09-09 06:17	2026-09-09 06:20	False	0.95	

... 13 more rows

... 2 more rows

- `DDN1_PNT_SL_SR01_NOZTEMP.PV` while `ST01-SR01` worked on the car (06:17–06:20): 44.3–45.2
- `DDN1_PNT_SL_SR02_FLOW.PV` while `ST01-SR02` worked on the car (06:17–06:20): 118.1–122.7

D3. All VINs with out-of-spec wheel-nut torque in the last week, the nutrunner used, and its last calibration work order

View: Traceability. Parameters: `from` = 2026-08-29 00:00:00, `to` = 2026-09-05 00:00:00.

```
MATCH (v:Vehicle)-[:hasTorqueResult]->(r:TorqueResult)-[:tightenedBy]->(tool:Tool)
WHERE r.joint_id STARTS WITH 'WHEEL' AND r.status = 'NOK' AND r.attempt = 1
  AND r.timestamp >= $from AND r.timestamp < $to
MATCH (tool)-[:partOf]->(ctrl:Component)-[:partOf]->(nr:Asset)
OPTIONAL MATCH (x:Asset)-[:hasWorkOrder]->(w:WorkOrder {wo_type: 'CAL'})
  WHERE x IN [nr, ctrl] AND w.created_at <= r.timestamp
WITH v, r, tool, nr, w ORDER BY w.created_at DESC
WITH v, r, tool, nr, collect(w)[0] AS last_cal
OPTIONAL MATCH (v)-[:hasTorqueResult]->(re:TorqueResult {joint_id: r.joint_id, attempt: 2})
RETURN v.vin AS vin, r.joint_id AS joint, r.torque_nm AS torque_nm, r.timestamp AS at, tool.tool_id AS
tool,
       nr.fl AS nutrunner, last_cal.wo_number AS last_calibration_wo, last_cal.created_at AS
calibrated_at,
       last_cal.short_text AS calibration_text, re IS NOT NULL AS re_hit
ORDER BY at
```

vin	joint	torque_nm	at	tool	nutrunner	last_calibration_wo	calibrated_at	calibration_text	re_hit
MDDA5XK0XD001391	WHEEL_LF	109.3	2026-09-03 14:07	TC14	DDN1-ASM-FN1-ST03-NR14				True
MDDA5XL01D001451	WHEEL_LF	109.9	2026-09-03 15:42	TC14	DDN1-ASM-FN1-ST03-NR14				True
… 13 more rows									

… 678 more rows

- 376 VINs, 693 NOK first rundowns, all on `TC14`; 38 were never re-hit.
- Run for the final week instead (8–15 Sep): 0 rows – the problem was fixed on 4 Sep.

D4. Spike in paint craters at topcoat inspection: upstream equipment via feeds, ranked by alarms and humidity excursions in the prior 2 hours

View: Quality. Parameters: `inspection` = DDN1-PNT-INSP-ST01, `from` = 2026-09-10 14:00:00, `to` = 2026-09-10 17:00:00.

```
MATCH (insp:Asset {fl: $inspection})<-[:feeds*1..3]-(u:Asset)
OPTIONAL MATCH (e:Asset)-[:partOf*0..2]->(u)
WITH DISTINCT u, e
OPTIONAL MATCH (al:Alarm)-[:raisedBy]->(e)
  WHERE al.event_time >= $from AND al.event_time <= $to AND NOT al.is_duplicate
OPTIONAL MATCH (e)-[:hasTimeSeries]->(t:TimeSeries) WHERE t.signal = 'RH' AND t.store_uri IS NOT NULL
RETURN u.fl AS upstream, count(DISTINCT al) AS alarms, collect(DISTINCT al.code) AS alarm_codes,
       collect(DISTINCT t.store_uri) AS humidity_series
ORDER BY alarms DESC
```

upstream	alarms	alarm_codes	humidity_minutes_over_75	score
DDN1-PNT-AHU-AHUTC2	184	H-07, H-15	90	274
DDN1-PNT-OV1	4	F-01, F-03, B-21, B-12	0	4
… 5 more rows				

- Crater spike: 3 craters in the hour from 10 Sep 16:00; window = 2 h before the spike hour to its end.
- Ranking = non-duplicate alarms on the upstream asset or anything partOf it + minutes of RH > 75 %. The AHU for booth B is first: humidity alarm flood and RH excursion, i.e. WO 4102963 (humidifier valve stuck open).

D5. All VINs that received parts from a suspect supplier lot (recall scope)

View: Traceability. Parameters: `lot` = WS-260904-B.

```
MATCH (l:SupplierLot {lot_id: $lot})<-[:fromLot]-(:Part)<-[c:containsPart]-(v:Vehicle)
OPTIONAL MATCH (v)-[:hasDefect]->(d:Defect) WHERE d.defect_code = 'WL-104'
WITH l, v, c, count(d) > 0 AS leaked
RETURN l.lot_id AS lot, l.supplier AS supplier, date(c.fitted_at) AS fitted_on, count(v) AS vehicles,
       sum(CASE WHEN leaked THEN 1 ELSE 0 END) AS with_door_leak
ORDER BY fitted_on
```

lot	supplier	fitted_on	vehicles	with_door_leak
WS-260904-B	Kumar Weather Seals Pvt Ltd	2026-09-04	293	0
WS-260904-B	Kumar Weather Seals Pvt Ltd	2026-09-08	593	31
… 2 more rows				

- Lot id scope: 1,785 vehicles. But the 4 Sep fitting session had 0 door leaks while 8–10 Sep had all of them (Part B finding: the lot id was not updated at T2-01).
- Refined recall scope (fitted from 8 Sep): 1,492 vehicles – the lot id alone would recall 293 good cars.

D6. Tags and work orders still not linked to any asset, and edges below 0.8 confidence by method

View: Maintenance. Parameters: none.

```
CALL () { MATCH (t:TimeSeries) WHERE NOT ()-[:hasTimeSeries]->(t) RETURN 'tag' AS kind, count(t) AS
unlinked,
          collect(t.tag)[..5] AS examples }
RETURN kind, unlinked, examples
UNION ALL
CALL () { MATCH (w:WorkOrder) WHERE NOT ()-[:hasWorkOrder]->(w) RETURN 'work order' AS kind, count(w) AS
unlinked,
          collect(w.wo_number + ' ' + w.short_text)[..5] AS examples }
RETURN kind, unlinked, examples
```

kind	unlinked	examples
tag	9	PRS.LP2.PR4.SPM, DD1_BIW_L1_ST01FX_CLMP1.PV, DDN1_BIW_L1R_09_A1TRQ.PV, DDN1_BIW_2L_R07_STATUS.PV, DDN1_BIW_L2_0R9_WELDCNT.PV
work order	122	4100865 Z3 oven profile run, within spec, 4101770 fan FAN vibration route check ok, 4101416 tip dresser ST07 tip dresser blade broken, replaced, 4100629 Station 04 L2 fixture locating pin worn, replaced, 4102142 robot no 2 L1 fault E-201, reset and restarted

- Outside the graph, waiting in the review app: workorder 122, document 44, tag 9.

D7. Plant head: how many cars built in these two weeks carry a known risk, and how many of them did inspection catch?

View: Quality. Parameters: `lot` = WS-260904-B, `lot_from` = 2026-09-08 00:00:00, `sealer` = DDN1-PNT-SL-ST02-SR03, `seal_from` = 2026-09-09 06:04:00, `seal_to` = 2026-09-09 10:18:00, `robot` = DDN1-BIW-L2-ST04-RB07, `weld_from` = 2026-09-07 13:17:00, `weld_to` = 2026-09-08 15:09:00.

```
CALL () {
  MATCH (v:Vehicle)-[:hasTorqueResult]->(r:TorqueResult {status: 'NOK', attempt: 1})
  WHERE r.joint_id STARTS WITH 'WHEEL'
    AND NOT EXISTS { (v)-[:hasTorqueResult]->(:TorqueResult {joint_id: r.joint_id, attempt: 2}) }
  RETURN DISTINCT v, 'wheel nut NOK, never re-hit' AS risk, ['AU-077'] AS catches
  UNION
  MATCH (v:Vehicle)-[c:containsPart]->(:Part)-[:fromLot]->(:SupplierLot {lot_id: $lot})
  WHERE c.fitted_at >= $lot_from
  RETURN DISTINCT v, 'door seal from suspect lot' AS risk, ['WL-104'] AS catches
  UNION
  MATCH (v:Vehicle)-[p:processedBy]->(:Asset {fl: $sealer}) WHERE p.start >= $seal_from AND p.start <=
$seal_to
  RETURN DISTINCT v, 'sealed while SR03 nozzle was cold' AS risk, ['WL-101'] AS catches
  UNION
  MATCH (v:Vehicle)-[p:processedBy]->(:Asset {fl: $robot}) WHERE p.start >= $weld_from AND p.start <=
$weld_to
  RETURN DISTINCT v, 'welded while R07 current was low' AS risk, ['BW-017'] AS catches
}
OPTIONAL MATCH (v)-[:hasDefect]->(d:Defect) WHERE d.defect_code IN catches
WITH risk, v, count(d) > 0 AS caught
RETURN risk, count(v) AS vehicles_at_risk, sum(CASE WHEN caught THEN 1 ELSE 0 END) AS
caught_by_inspection,
       sum(CASE WHEN caught THEN 0 ELSE 1 END) AS not_caught
ORDER BY vehicles_at_risk DESC
```

risk	vehicles_at_risk	caught_by_inspection	not_caught
door seal from suspect lot	1492	85	1407
welded while R07 current was low	647	103	544
… 2 more rows			

- Windows are the out-of-spec periods found in Part B (re-computed from the time series).
- `not_caught` is the field-risk number: cars that were exposed but never showed the defect at inspection. For the cold-weld window that is expected (only some welds are undersize and the chisel test samples); for the door-seal lot it means the shower test did not see every bad seal.

Appendix B – GraphRAG test questions: expected vs actual answers

From `reports/14_graphrag.md` (mode: openai). Answers are cut to 95 characters; the full answers and the Cypher behind each one are in that file. “Pass” is the automatic check; see section 4 for the one held-out answer that fails on manual review.

Evaluation set (10, incl. 2 “I don’t know”)

#	Question	Expected	Answer	Pass
1	Why did water-leak defects rise on Line 2 last Tuesday?	WS-260904-B	On 2026-09-08, there were 28 defects with defect code WL-104 on Line 2, associated with …	yes
2	Which robot caused the cold welds on 7 and 8 September, and which work order fixed it?	DDN1-BIW-L2-ST04-RB07, 4102961	The robot DDN1-BIW-L2-ST04-RB07 caused the cold welds on 7 and 8 September 2026. …	yes
3	What caused the paint craters on 10 September?	DDN1-PNT-AHU-AHUTC2, AN-006	The paint craters on 10 September were caused by an anomaly AN-006 on DDN1-PNT-AHU-AHUTC2. …	yes
4	Which nutrunner caused the low wheel-nut torque defects?	DDN1-ASM-FN1-ST03-NR14, AN-002	The nutrunner DDN1-ASM-FN1-ST03-NR14 caused 38 low wheel-nut torque defects. …	yes
5	How many vehicles received parts from supplier lot WS-260904-B?	1785	1785 vehicles received parts from supplier lot WS-260904-B. [WS-260904-B, 1785]	yes
6	Was the Oven 2 zone 3 over-temperature on 12 September a quality problem?	AN-008, false_alarm	The over-temperature on 12 September in DDN1-PNT-OV2-Z3 was judged as a false_alarm. [AN-008] …	yes
7	Which sealer robots processed vehicle MDDA5XK0XD004375 in sealer station 2?	DDN1-PNT-SL-ST02-SR03, DDN1-PNT-SL-ST02-SR04	Vehicle MDDA5XK0XD004375 was processed by sealer robot DDN1-PNT-SL-ST02-SR03 on 2026-09-09 …	yes
8	List the temperature sensors in Topcoat Oven 2.	PaintShop-OVN2-Z3-Temp	The temperature sensor in DDN1-PNT-OV2-Z1 is tagged as PaintShop-OVN2-Z1-Temp and measures in …	yes
9	What was the paint film thickness on vehicle MDDA5XK0XD004375?	I don’t know	I don’t know. The schema does not include time-series data or specific measurements like paint …	yes
10	Who was the shift supervisor on 9 September?	I don’t know	I don’t know. The schema does not include any data about people, roles, or shift supervisors.	yes

Held-out round 2 (unseen)

#	Question	Expected	Answer	Pass
1	Which anomaly was judged an equipment problem rather than a quality problem, and on which asset?	AN-005, DDN1-PNT-AHU-AHUTC1-SF	The anomaly AN-005 was judged an equipment problem rather than a quality problem on the asset …	yes
2	How many defects are attributed to anomaly AN-004?	36	There are 36 defects attributed to anomaly AN-004. [AN-004]	yes
3	Which work order fixed the sealer robot nozzle problem?	4102962	The work order 4102593 fixed the sealer robot nozzle problem on DDN1-PNT-SL-ST01-SR01 by …	yes
4	How many vehicles got a door seal from lot WS-260903-B?	299	299 vehicles received a door seal from lot WS-260903-B. [WS-260903-B]	yes
5	What is the warranty claim rate for Line 2?	I don’t know	I don’t know. The schema does not include data related to warranty claims or their rates. …	yes